ASSEMBLY INSTRUCTIONS

FULLY ASSEMBLED SENSORS

LIST OF COMPONENTS

- Assembled and pre-flashed PCB
- Display
- SPS30 – Particulate Matter sensor
- SPS30 connector cable
- SD card
- Battery
- GPS antenna
- Sensor box
- Thick double-sided tape
- 4x M3 hexagonal spacers
- 4x M3 screws
- Thin double-sided tape
- Garmin mount
- Sticker

LIST OF TOOLS

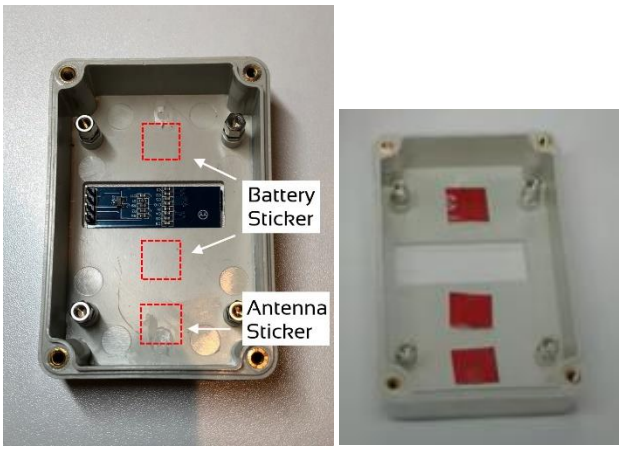
- M3 screwdriver
- Scissors
- M3 hexagonal nut drive

ASSEMBLY STEPS

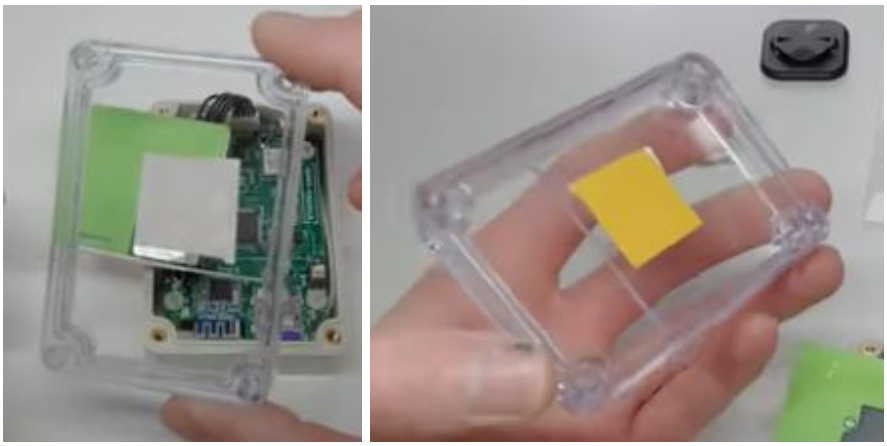
Follow the instructions according to this video:
https://www.youtube.com/watch?v=Utk1S457eWM&ab_channel=openseneca

Prior to the instructions of the video, there are a couple more steps to do:

1. Screw the 4 hexagonal spacers into the sensor box mounting holes with the help of a nut driver (see picture below for reference). Note: the holes in the sensor box are not threaded, the thread will be created when screwing the spacer into the plastic.
2. Cut three 1cmx1cm pieces of the thick double-sided tape, peel one side only, and place them on the bottom of the sensor box as follows (see picture). Note: the box needs to be in a specific orientation, with the display hole on the left side.



3. Cut a 2cmx2cm piece of the thin double-sided tape and place it on the inner part of the lid. Note: when looking at the box lid from the outside (like if the box was closed) and with the opening at the bottom, the double-sided tape needs to be placed on the right side, leaving around 2mm of space between the tape and the box opening. Therefore, when looking at it from the inner side of the box lid, the double-sided tape is placed on the left side.



4. Flash PCB with the latest firmware.
5. Now everything is ready to start following the instructions in the video step by step.

TESTING STEPS

In order to make sure all the PCBs are working properly – as there is an error rate in the PCBA production – some verifications can be done:

1. When the PCB is powered by micro-USB, a red LED next to the USB connector turns ON.
2. When the PCB is powered (by battery or USB), the RGB LED turns on and starts blinking. Other components also turn on:
    a. The SPS30 particulate matter sensor (green sensor) has a fan that starts turning – you can hear the fan.
    b. The display turns on and shows “open-seneca” and then starts showing values
3. The STAT LED (next to the SD card slot) lights if there is anything wrong with your board. The colours mean the following:
    a. - red: battery voltage reading not between 2.6V and 5V
    b. - blue: humidity reading not between 0 and 100% (faulty SHT sensor?)
    c. - purple: SD card is not found or cannot be read/written
    d. - cyan: SPS sensor does not respond
4. If the SD card is not recognised by the PCB, the display shows “No SD”. If the SD is correctly mounted in the SD card holder, then the problem could be that the SD card holder is not soldered properly.
5. If the computer does not recognised the sensor via USB with the right drivers (stm32 driver), the USB is not soldered properly.
6. On the display, the indicated values are temperature (left) and humidity (right, going from 0-99%). If the values shown do not make sense for ambient temperature and humidity, it means the temperature and humidity sensor is not soldered properly.

7. When outdoors, a blue light will start blinking when GPS connection is found.
8. Testing for the charging circuit: plug sensor via USB overnight and check in the morning that the battery is charged (full icon on the display).